Name: _____



New York State Testing Program

Elementary-level Science Test

Grade 5

Spring 2026

RELEASED QUESTIONS

Elementary-level Science Test

TIPS FOR TAKING THE TEST

Here are some ideas to help you do your best:

- Be sure to read all the directions carefully.
- Read each question carefully.
- Think about the answer before making your choice or writing your answer.
- Make sure you read all the information given for each question.
- Diagrams included in the test are not drawn to scale unless otherwise noted.
- You have a ruler and a calculator that you can use on the test if they help you answer the question.

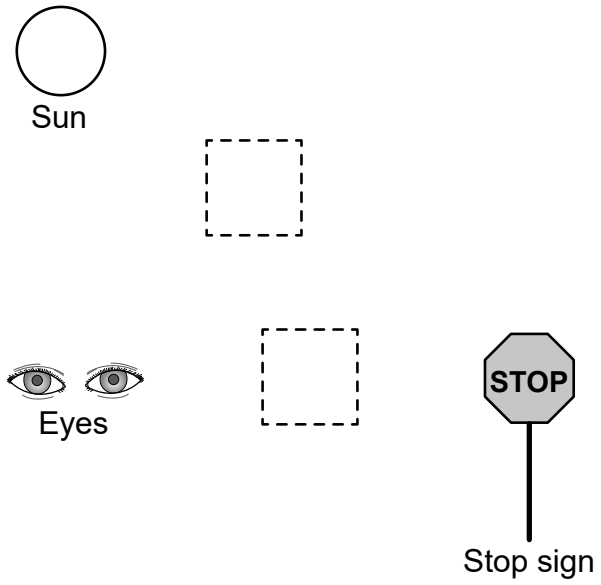
Base your answers to questions 1 through 4 on the information below and on your knowledge of science.

Sunlight and Starlight

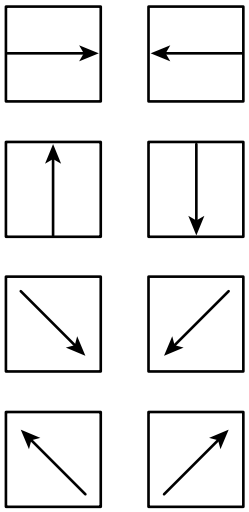
Our solar system exists in the Milky Way Galaxy. There are about 100 billion stars in the Milky Way and the Sun is one of them. The Sun is the only star in our solar system. The Sun is responsible for many patterns and events that occur on Earth.

1 Sunlight allows objects to be visible on Earth. From the choices, place one arrow in *each* box to complete the model to show the correct path of light that permits a stop sign to be seen. [1]

Path of Light Model



Arrow Choices:

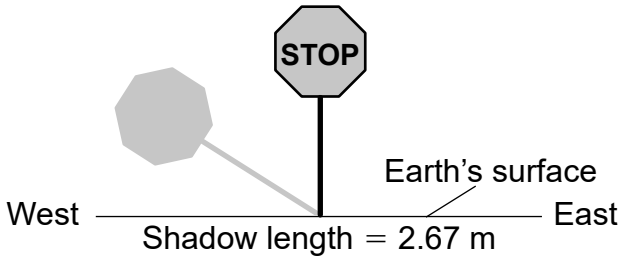


GO ON

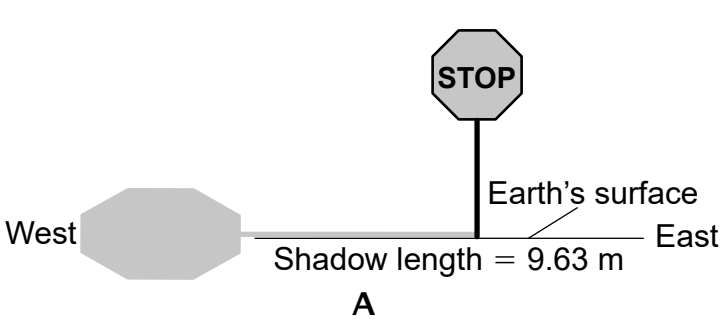
The data below shows some information about the shadow formed by sunlight for a 2.0-meter tall stop sign in Rochester, New York on September 23, 2023.

Time of Day	**Length of Stop Sign Shadow** (m)
8:15 a.m.	8.63
10:45 a.m.	2.67
1:15 p.m.	1.89
3:45 p.m.	3.02
6:15 p.m.	13.21
8:45 p.m.	None

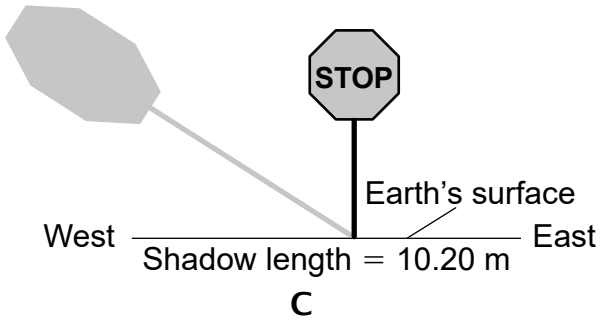
The direction and length of the shadow of the 2.0-meter stop sign for 10:45 a.m. are represented in the diagram below.



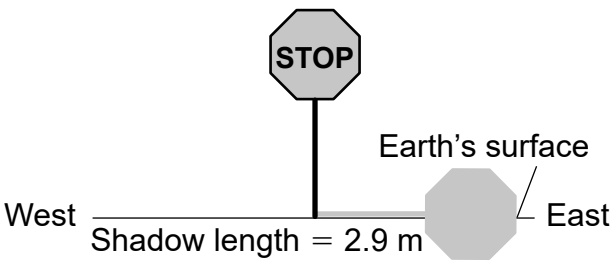
2 Which diagram best represents the direction and length of the same stop sign's shadow at 5:15 p.m.?



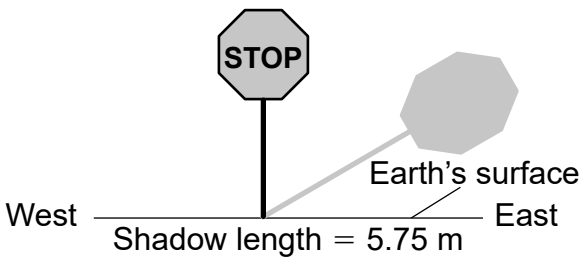
A



C



B



D

GO ON

The data table below shows some information about sunrise and sunset in Rochester, New York on September 23, 2023.

	Time of Day
Sunrise	6:58 a.m.
Sunset	7:07 p.m.

3 Which table correctly identifies the amount of daylight (hours and minutes) in Rochester, New York on December 21, 2023 and June 20, 2024?

A

Month	Daylight
December	8 h 59 min
June	15 h 24 min

B

Month	Daylight
December	15 h 24 min
June	8 h 59 min

C

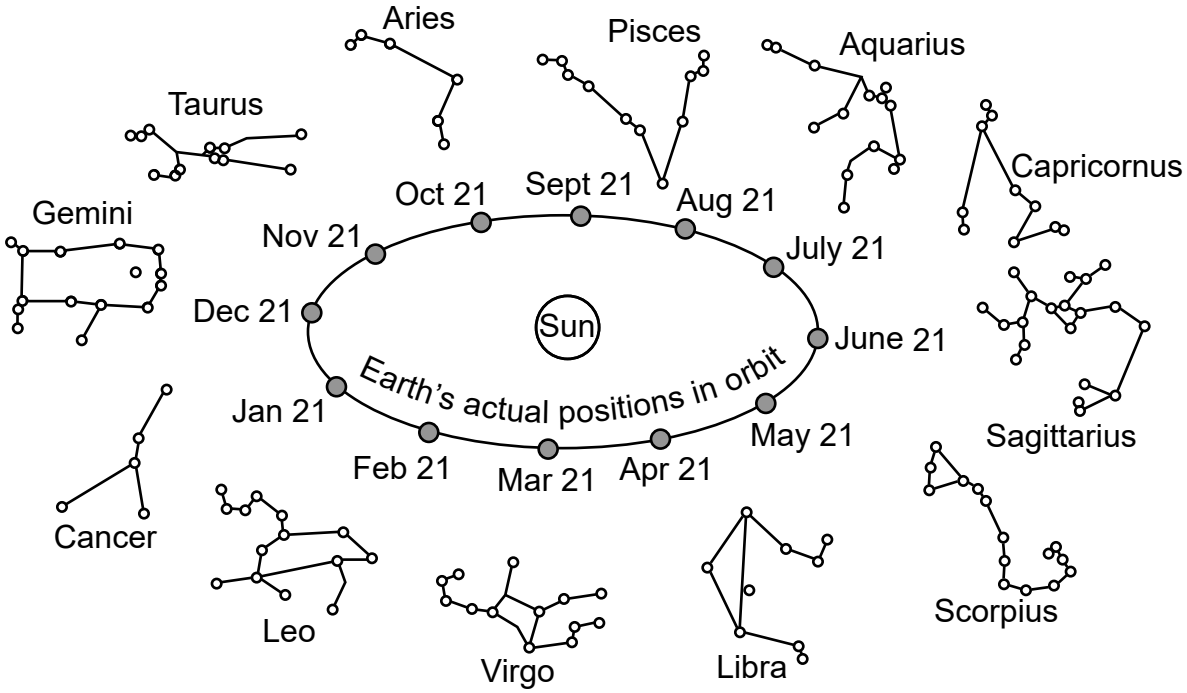
Month	Daylight
December	12 h 9 min
June	11 h 51 min

D

Month	Daylight
December	11 h 51 min
June	12 h 9 min

After the Sun sets, some stars are visible in the night sky. The model below shows different groups of stars (constellations) that are visible in the night sky in Rochester, New York as Earth orbits the Sun.

Model of Constellations



4 From the choices below, place **one** correct constellation name in *each* box to indicate the constellations that would be entirely visible and would ***not*** be visible from Rochester, New York at night on March 21. [1]

Choices for Constellations:

- Gemini
- Libra
- Pisces
- Sagittarius
- Virgo

Visible [ ]

Not Visible [ ]

GO ON

Base your answers to questions 5 through 9 on the information below and on your knowledge of science.

Minerals of New York State

A variety of minerals are found in New York State. Many minerals are mined and taken out of rock formations. They are used in industry, construction, computers, medicine, and jewelry.

During a unit of study, a teacher gave students five different mineral samples that can be found in New York State. The students used appropriate safety equipment and followed safety procedures as they made observations and performed tests on these minerals to determine their properties. Their observations are shown in the table below.

Minerals of New York State and Their Observed Properties

Mineral	Color	Streak (color in powdered form)	Hardness	Surface Reflectivity	Crystal Shape
calcite	clear/many colors	clear/white	3	non-metallic	cubic
garnet	red-brown	clear	7	non-metallic	cubic
halite	clear/white/ gray	clear/white	2-2.5	non-metallic	cubic
hematite	red-brown	red-brown	5-6.5	non-metallic	hexagonal
talc	pearly white	white	1	non-metallic	not observable

5 Which mineral observations could *both* be used to determine that a sample of talc is different from calcite or halite?

A hardness and crystal shape

B streak and crystal shape

C hardness and surface reflectivity

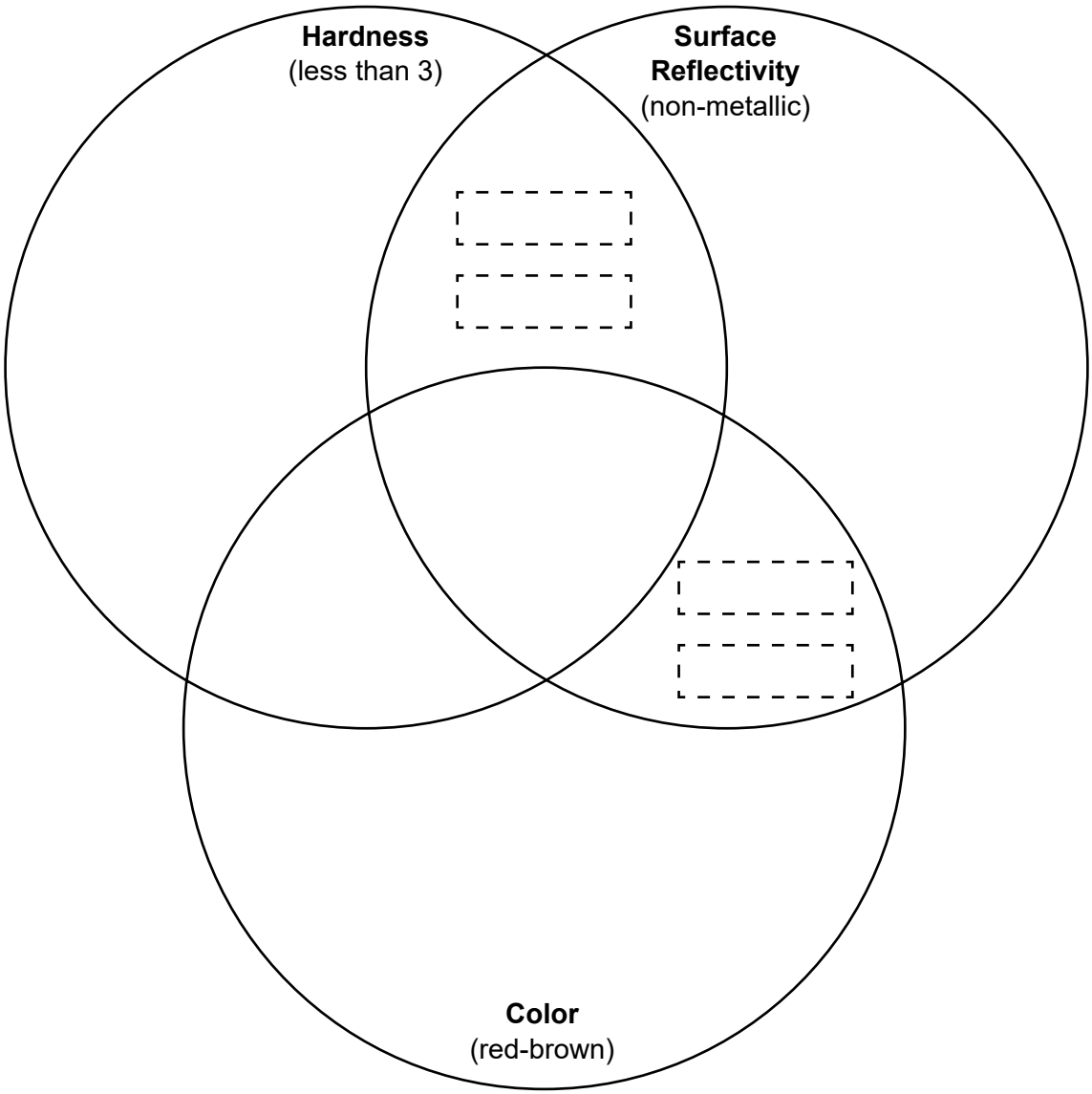
D streak and surface reflectivity

GO ON

6 The model represents how three properties can be used to classify these minerals. The areas of overlap show shared properties. Based on the properties shown in *Minerals of New York State and Their Observed Properties*, place the names of **four** of the minerals from the choices below in the correct boxes within the model. [1]

Minerals:

Hematite | Garnet | Talc | Halite | Calcite

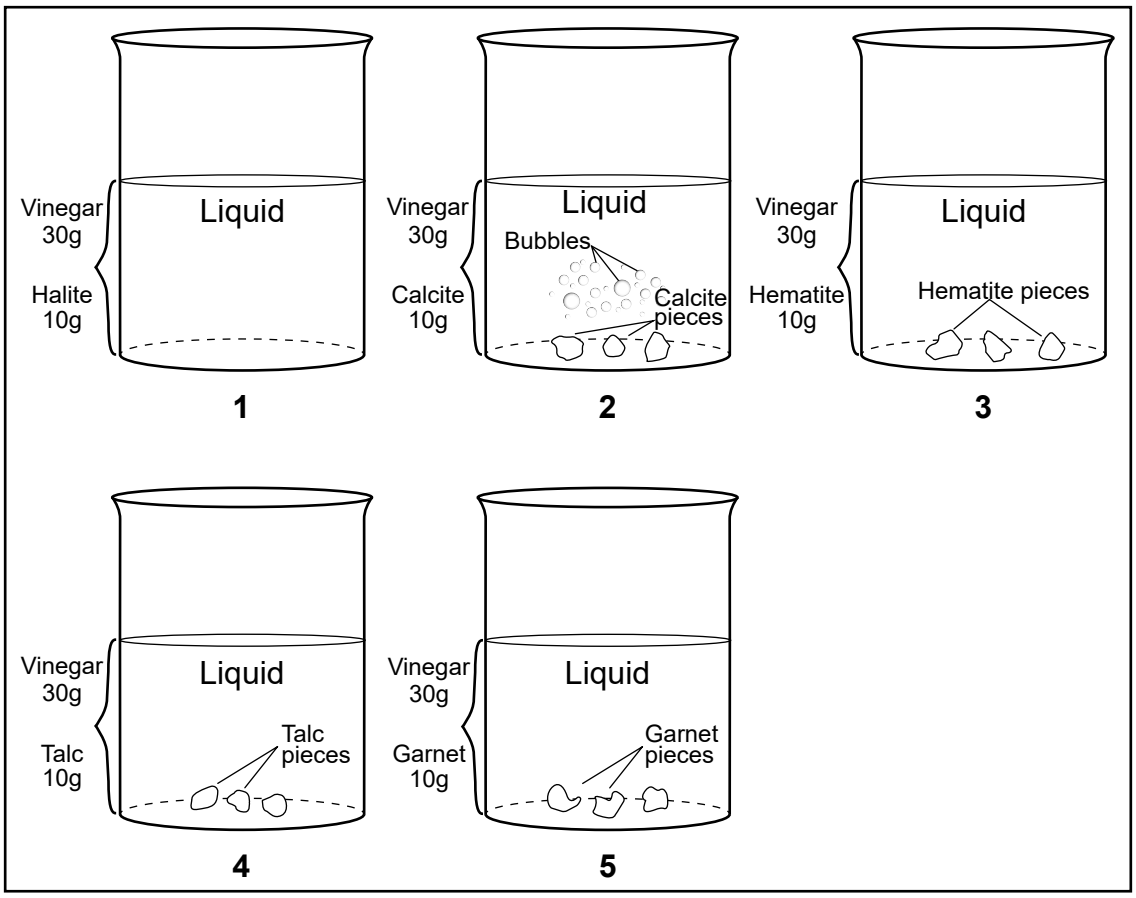


GO ON

The students then investigated how vinegar affects these minerals. They placed equal amounts of each mineral into beakers with equal amounts of vinegar.

The students made a model, shown below, of the results of the investigation.

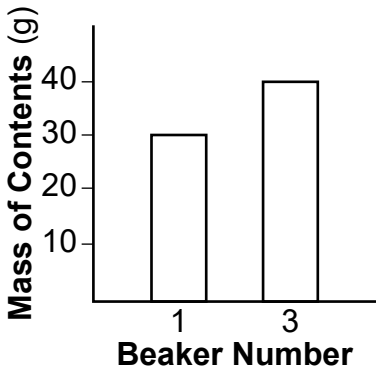
Model of Investigation Results



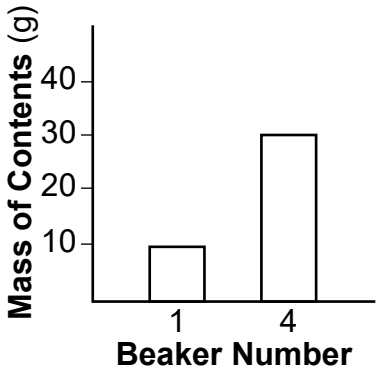
GO ON

7 Which graph correctly shows the total mass of the contents of two of the beakers in the *Model of Investigation Results*?

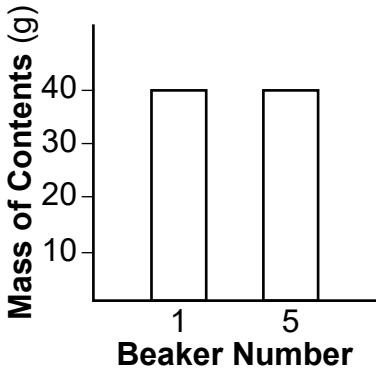
A



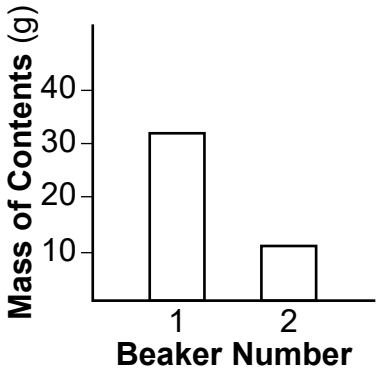
C



B



D



8 A student claimed that one of the beakers showed evidence that the mixing of substances resulted in a new substance. Identify the beaker number in which a new substance was formed. Then identify the evidence in the model that supports your choice. [1]

Beaker number: _____

Evidence in model: _____

_____

_____

9 Which piece of evidence correctly identifies an example of solubility?

A The halite is not visible in the vinegar.

B The calcite pieces settled to the bottom of the beaker.

C The hematite pieces became more rounded.

D The garnet pieces had an increase in reflectivity.

GO ON

Base your answers to questions 10 through 13 on the information below and on your knowledge of science.

Wolves in Yellowstone National Park

In the 1800s, wolves were commonly found from the cold Arctic regions to Mexico. By the early 1900s, almost no wolves were found in the United States because of habitat loss and hunting by humans.

In 1973, the northern Rocky Mountain wolf was listed as an endangered species. As a solution, wolves were reintroduced into the Greater Yellowstone Ecosystem. From 1995 to 1997, 41 wild wolves from other areas were released into Yellowstone National Park.

Wolves are highly social animals. They eat a wide variety of prey, both large and small. They mate in February, and females give birth to an average of five pups in April.

At the end of 2022, there were 108 wolves in Yellowstone. These wolves lived in 10 different packs, ranging in size from 4 to 25 wolves.

10 Which fact does ***not*** support the argument that being part of a wolf pack helps members survive?

A The leading cause of death for wolves outside Yellowstone Park is humans.

B Food and protection is provided to newborn pups.

C Large prey are hunted and killed by wolves.

D Regions in which wolves live are defended from other wolves.

GO ON

In Yellowstone, about half of the wolves are black in color, and the other half are gray in color.

Studies of these wolves have shown the following:

- Gray-colored wolves are more aggressive than black-colored wolves during encounters with other wolf packs.
- The litter sizes of gray-colored wolves are slightly larger than the litter sizes of black-colored wolves.
- Wolves often choose mates of the opposite color.
- Black-colored wolves have a greater survival rate from some diseases than gray-colored wolves.

Gray-colored and Black-colored Wolves in Yellowstone



11 Which statement correctly summarizes what the studies found comparing wolves with different coat colors?

A Wolves have traits that can be influenced by the environment.

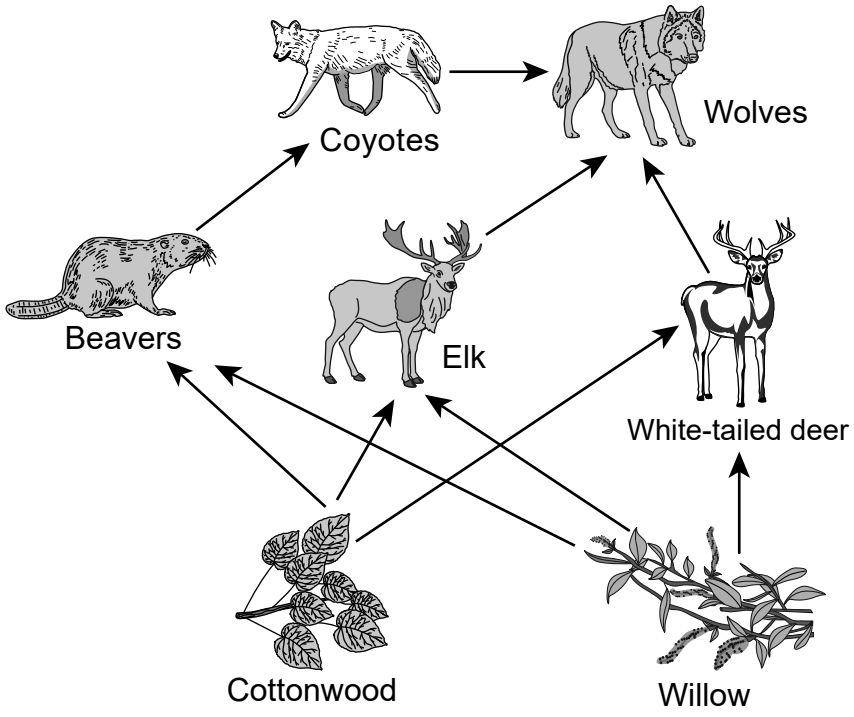
B Variations in characteristics of wolves may provide advantages in surviving and reproducing.

C Traits of wolves are inherited from parents, and variations of these traits exist in a pack.

D Internal structures of wolves allow for survival, growth, and successful reproduction.

The incomplete food web model below shows some information about feeding relationships in Yellowstone National Park.

Yellowstone National Park Food Web



12 Identify **one** food source for the white-tailed deer in Yellowstone National Park. Then, describe how this food source would be affected if the white-tailed deer population changed when more wolves were introduced into the park. Include a reason for this change in your description. [1]

Food source: _____

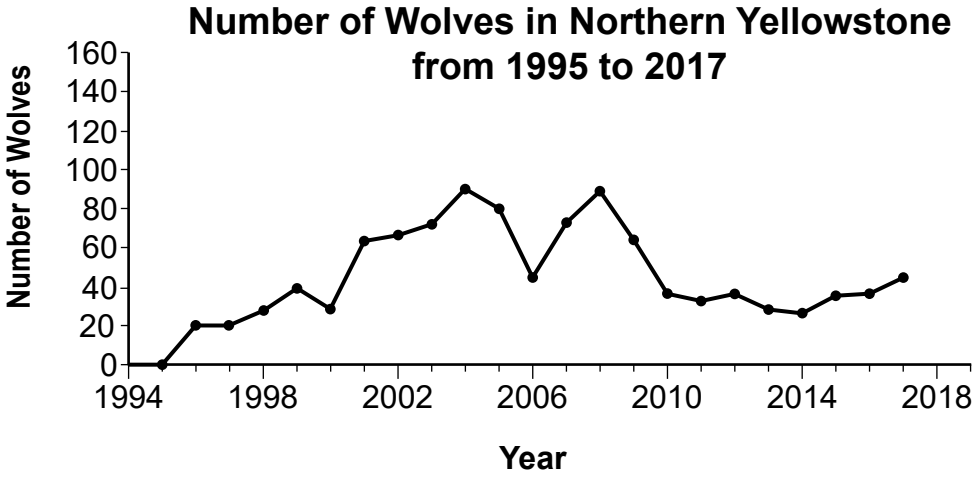
Effect on food source: _____

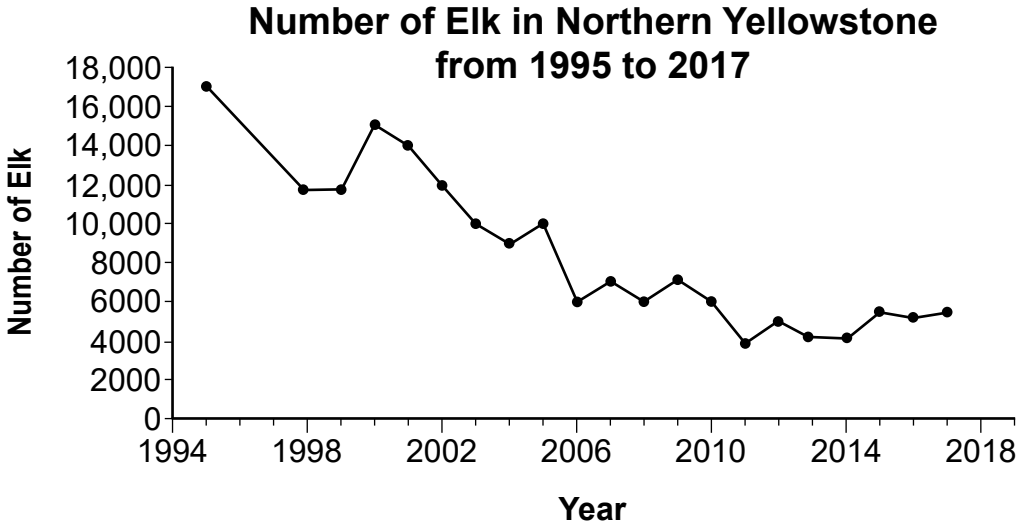
_____

_____

GO ON

The graphs below show some information about the number of wolves and elk found in the northern range of Yellowstone National Park since wolves were reintroduced to the park in 1995.





13 Which piece of evidence from the graphs supports the claim that wolves are dependent on elk for their survival in Yellowstone National Park?

A From 2010 to 2017, both the wolf and elk populations decreased, then increased.

B Both the wolf and elk populations increased from 2005 to 2006.

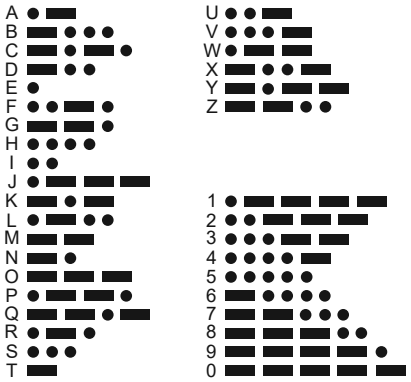
C From 2000 to 2004, the number of wolves increased while the number of elk decreased.

D The highest wolf populations were in 2004 and 2008, while the highest elk populations were in 1995 and 2000.

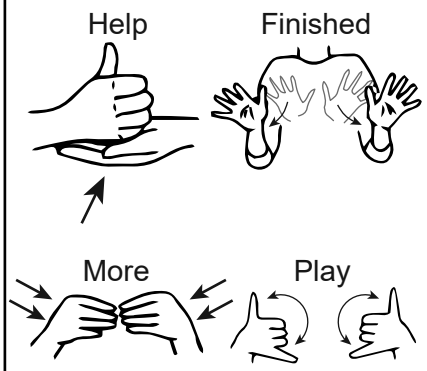
Base your answers to questions 14 through 18 on the information below and on your knowledge of science.

Information Transfer

People communicate in order to share information with others, express needs, and maintain friendships. Throughout history, people around the world have created and used different methods of communication. The table below describes a few of these methods.

Method	Description	Example
Drum signals	System that uses different sound combinations on a drum.	
Morse code	System that uses long and short sounds, long or short flashing lights, or dots and dashes, to represent letters and numbers.	
String-and-can telephone	System that allows a person to talk into a can on one end and be heard through the can on the other end.	

(Table continued)

<table>
<tr><th>Method</th><th>Description</th><th>Example</th></tr>
<tr><td>Sign language</td><td>System that uses hand movements to represent letters and words.</td><td> 
</td></tr>
<tr><td>Binary code</td><td>System that uses 0 and 1 to represent letters, numbers, and symbols used in computers and cell phones.</td><td>
<table>
<tr><th>Number</th><th>Binary Code</th></tr>
<tr><td>0</td><td>0</td></tr>
<tr><td>1</td><td>01</td></tr>
<tr><td>2</td><td>10</td></tr>
<tr><td>3</td><td>11</td></tr>
<tr><td>4</td><td>100</td></tr>
<tr><td>5</td><td>101</td></tr>
<tr><td>6</td><td>110</td></tr>
<tr><td>7</td><td>111</td></tr>
<tr><td>8</td><td>1000</td></tr>
<tr><td>9</td><td>1001</td></tr>
<tr><td>10</td><td>1010</td></tr>
<tr><td>11</td><td>1011</td></tr>
<tr><td>12</td><td>1100</td></tr>
<tr><td>13</td><td>1101</td></tr>
<tr><td>14</td><td>1110</td></tr>
<tr><td>15</td><td>1111</td></tr>
</table>
<table>
<tr><th>Number</th><th>Binary Code</th></tr>
<tr><td>16</td><td>10000</td></tr>
<tr><td>17</td><td>10001</td></tr>
<tr><td>18</td><td>10010</td></tr>
<tr><td>19</td><td>10011</td></tr>
<tr><td>20</td><td>10100</td></tr>
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<tr><td>23</td><td>10111</td></tr>
<tr><td>24</td><td>11000</td></tr>
<tr><td>25</td><td>11001</td></tr>
<tr><td>26</td><td>11010</td></tr>
<tr><td>27</td><td>11011</td></tr>
<tr><td>28</td><td>11100</td></tr>
<tr><td>29</td><td>11101</td></tr>
<tr><td>30</td><td>11110</td></tr>
<tr><td>31</td><td>11111</td></tr>
</table>
</td></tr>
</table>

14 Identify **two** communication methods that use a pattern in order to transfer information. Use evidence from the table to support each choice. [1]

Method 1: _____

Evidence: _____

_____

Method 2: _____

Evidence: _____

_____

GO ON

15 Using the information in the table, which statement about energy using the string-and-can telephone is most accurate?

A Sound energy is transferred through the string.

B Electrical energy is transferred through the string.

C Sound energy is converted into electrical energy in the string.

D Heat energy is converted into sound energy in the can.

16 When drum signals are used for communication, what happens to the energy of the mallet, drum, and surrounding air when the mallet hits the drum?

A The mallet gains energy from the drum and the surrounding air.

B The mallet and drum both gain energy from the surrounding air.

C The mallet loses energy, and this energy is transferred to the drum, but not to the surrounding air.

D The mallet loses energy, and this energy is transferred to the drum and then to the surrounding air.

Two friends live in buildings across the street from each other. They want to build their own method of communication that would allow them to find out if the other is available to meet outside. They can see and hear each other from their bedrooms. In order to be effective, this method must meet the following criteria and constraints.

Criteria:

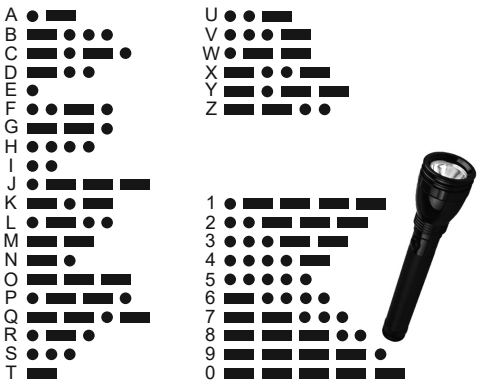
- be able to transfer information easily from one person to another
- inexpensive
- can be used either day or night

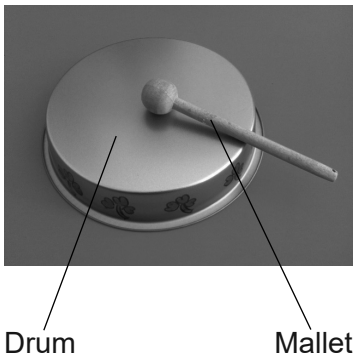
Constraints:

- must be quiet after 7:00 p.m.
- must be kept in their bedrooms
- must not take up a lot of space

They have proposed two different methods:

Method 1	**Method 2**
Morse code and flashlight	Drum signals





17 Identify the method that allows the friends to *most* effectively receive each other's messages, and explain how this method best meets the criteria *and* constraints. [1]

Method: _____

Explanation: _____

_____

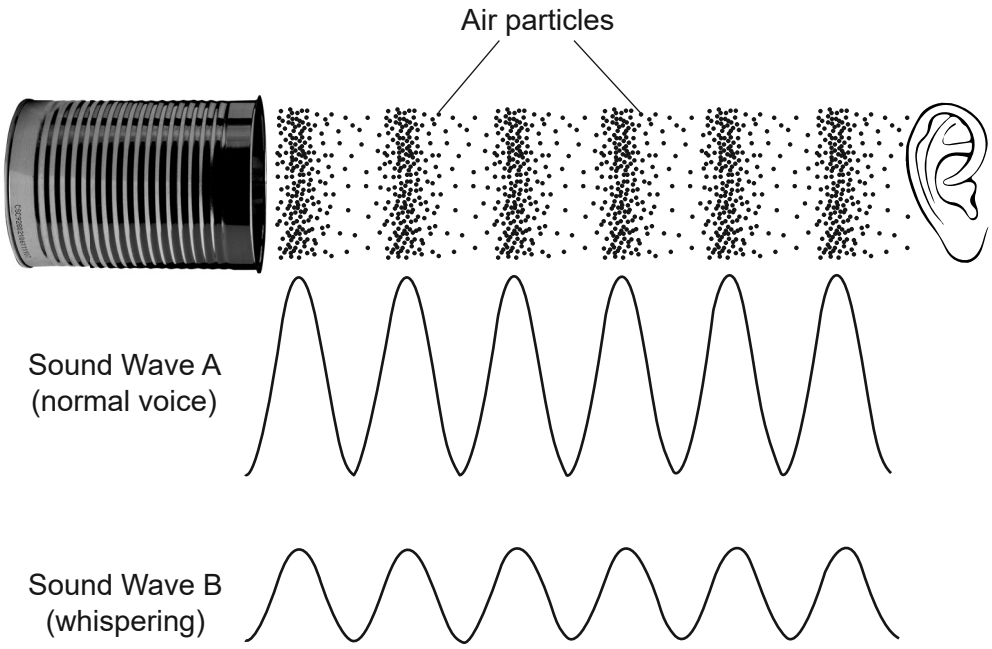
_____

GO ON

Air particles vibrate as a sound wave passes through them. The model below represents the motion of air particles caused by sound waves leaving a can and being heard by a person. The patterns of the particles can be changed into a model shaped like a wave.

Wave *A* represents a student speaking with their normal voice volume. Wave *B* represents the same student whispering. In both examples, the student spoke using the same pitch.

String-and-Can Telephone Waves



18 Compare the amplitude and wavelength of Wave *A* to the amplitude and wavelength of Wave *B*. Mention both waves in your responses. [1]

Amplitude: _____

_____

Wavelength: _____

_____

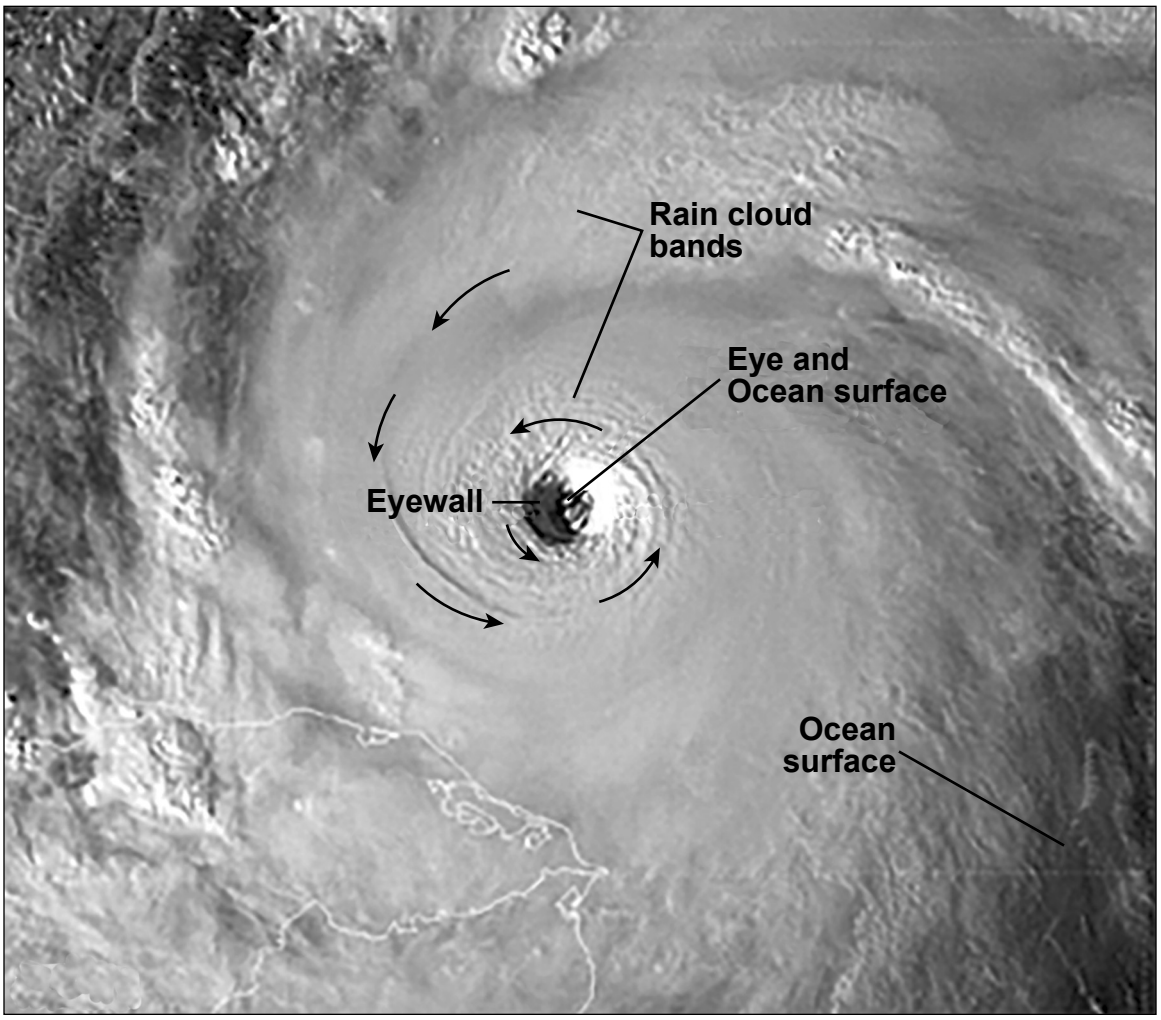
GO ON

Base your answers to questions 24 through 27 on the information below and on your knowledge of science.

Hurricane Formation

Hurricanes form from smaller storms. Heat energy in warm ocean water causes water vapor to enter the atmosphere. This vapor rises and cools, creating a storm system that forms clouds. Water falls back to Earth as rain. The storm starts to spin due to Earth's rotation. For storms to strengthen, there needs to be a continuous supply of warm ocean water. The storm spins around a center, called the eye. The winds in a hurricane are 74 miles per hour (mph) or greater. The strongest winds and thunderstorms occur at the edge of the eye, called the eyewall. Bands of rain clouds circle around the eye.

The photograph below shows a satellite image of a hurricane. Some parts of a hurricane are labeled. Arrows show the pattern of winds around the eye.



GO ON

24 The table below describes water cycle processes that are the cause of or the effect of hurricane formation. Two processes are labeled 1 and 2.

Process	Description in Hurricane Formation
1	Heat energy in warm ocean water causes water vapor to enter the atmosphere.
2	Water vapor cools as it rises in the air, forming clouds.
precipitation	Rain falls back to Earth in bands of storms around the eye of the hurricane.
runoff	Water on Earth's surface from thunderstorms in hurricanes moves into lakes and rivers.

Which set of water cycle processes correctly completes the table?

A 1-groundwater flow 2-condensation

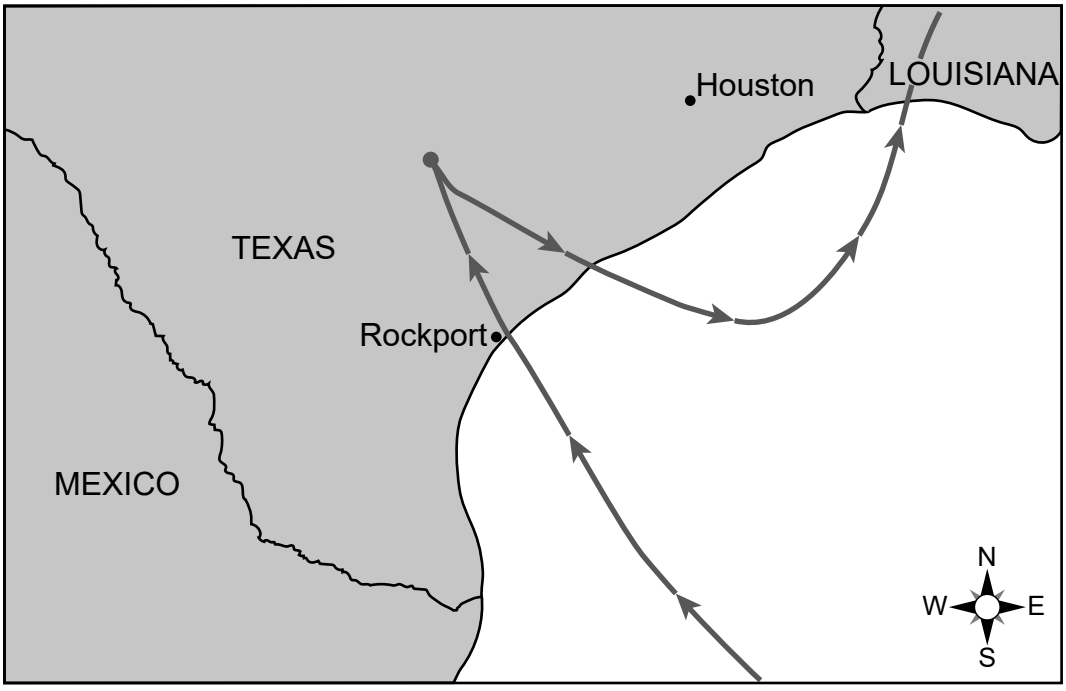
B 1-evaporation 2-groundwater flow

C 1-condensation 2-evaporation

D 1-evaporation 2-condensation

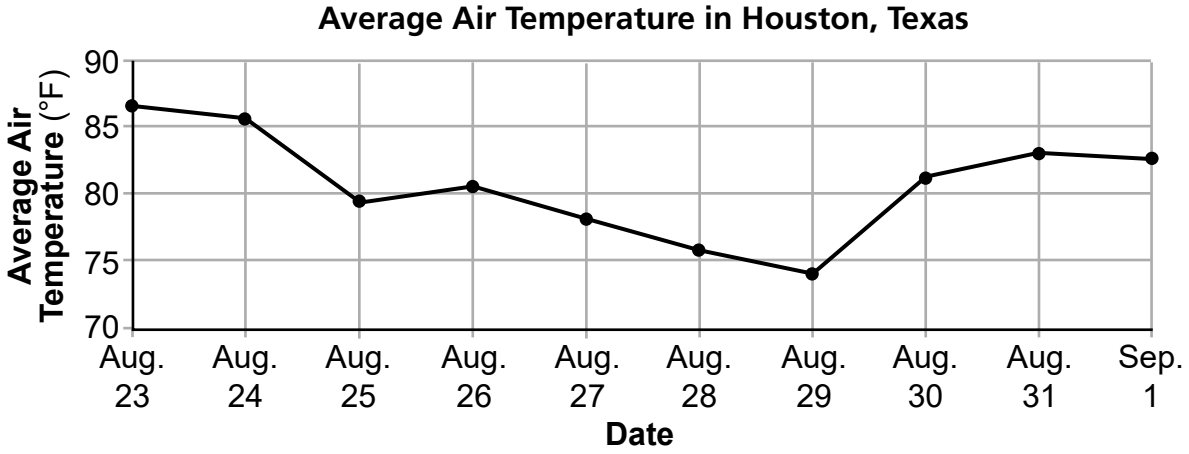
Hurricane Harvey was a strong hurricane that made landfall on August 25, 2017 near Rockport, Texas. The map below shows the path of the hurricane as it moved toward central Texas and then moved east over a body of water. Arrows show the path of the hurricane. Harvey remained within Texas borders for 60 hours.

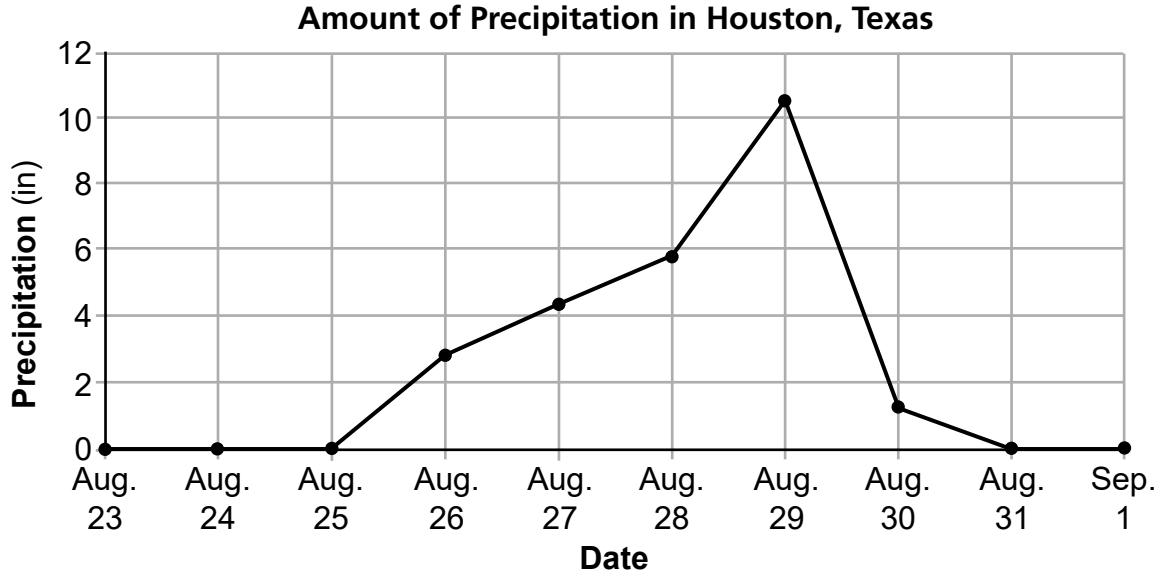
Path of Hurricane Harvey in Texas

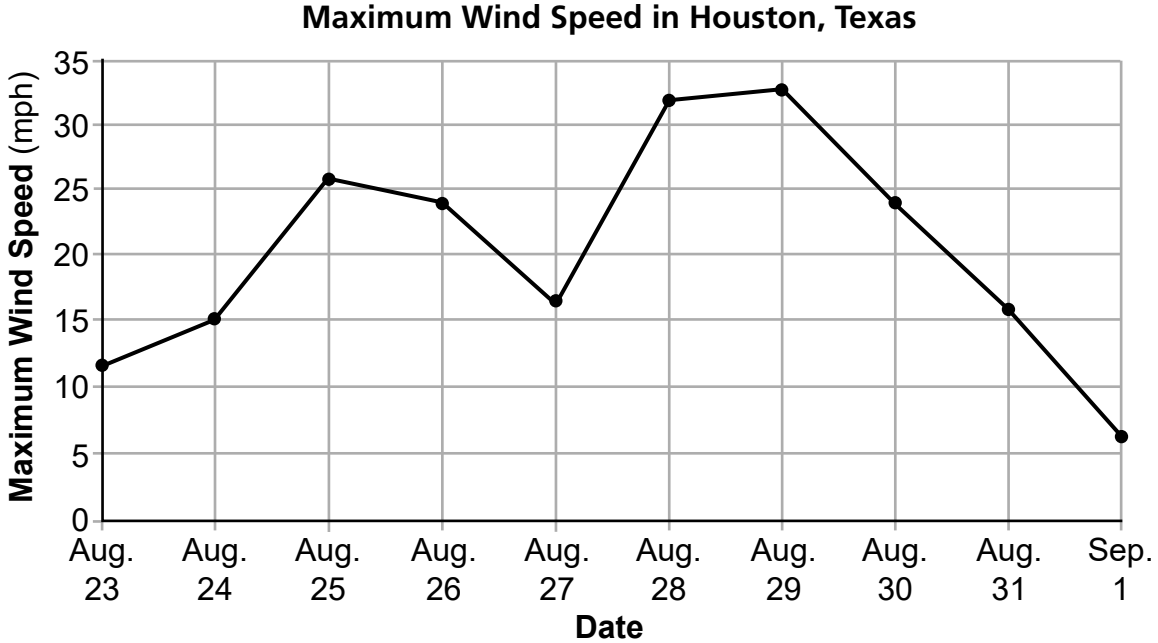


GO ON

The graphs show some information about weather conditions in Houston, Texas in the last week in August, 2017.







GO ON

25 Based on the information in the graphs, select **one** date from the table during which the greatest effects of Hurricane Harvey in Houston, Texas were recorded.

Date	
August 27, 2017	
August 28, 2017	
August 29, 2017	

Then, complete the passage below by placing the correct term in *each* box from the choices given. Terms may be used more than once. [1]

Term Choices:

decrease	increase	remain constant

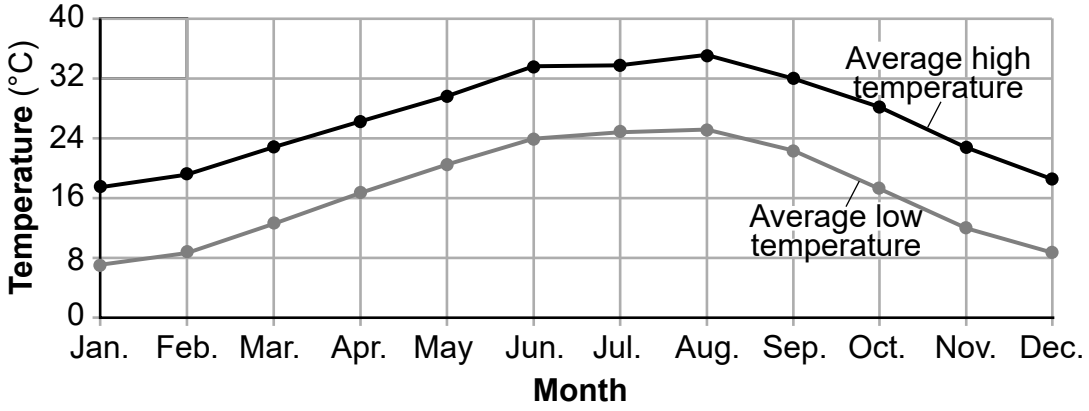
Passage

Based on the patterns seen in the graphs, Hurricane Harvey caused the air temperature to [ ], the precipitation to [ ], and the wind speed to [ ].

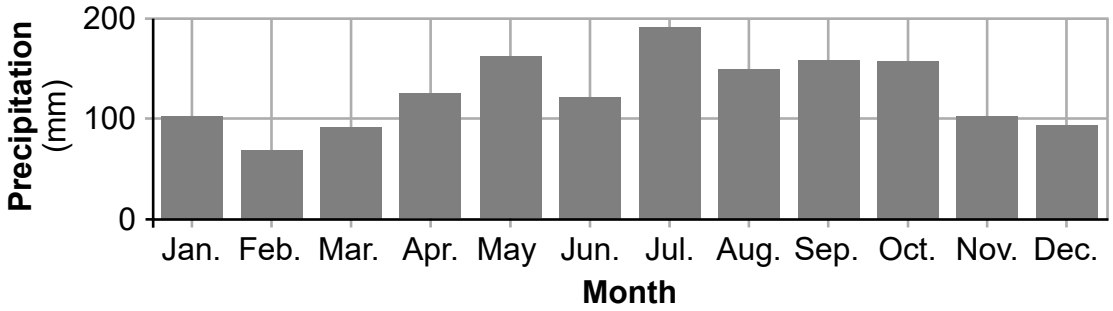
Hurricanes usually develop in the southern regions of the Atlantic Ocean instead of the northern regions of the Atlantic Ocean.

The graphs and table below show some information about the climate of Houston, Texas.

Monthly Average High and Low Air Temperatures in Houston, Texas



Monthly Average Precipitation in Houston, Texas



Monthly Average Surface Water Temperature of the Body of Water near Houston, Texas

Month	Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.
Temperature (°F)	55.9	57.6	65.0	71.3	77.7	83.5	85.9	86.4	83.3	76.0	65.8	58.8

26 Hurricanes strengthen over the body of water near Houston, Texas. Based on the information provided, which statement correctly describes the reason this happens?

A In late summer months, the increase in precipitation near Houston supplies energy to further develop storms into hurricanes.

B In late summer months, the surface water temperature increases enough to supply energy for the development of hurricanes.

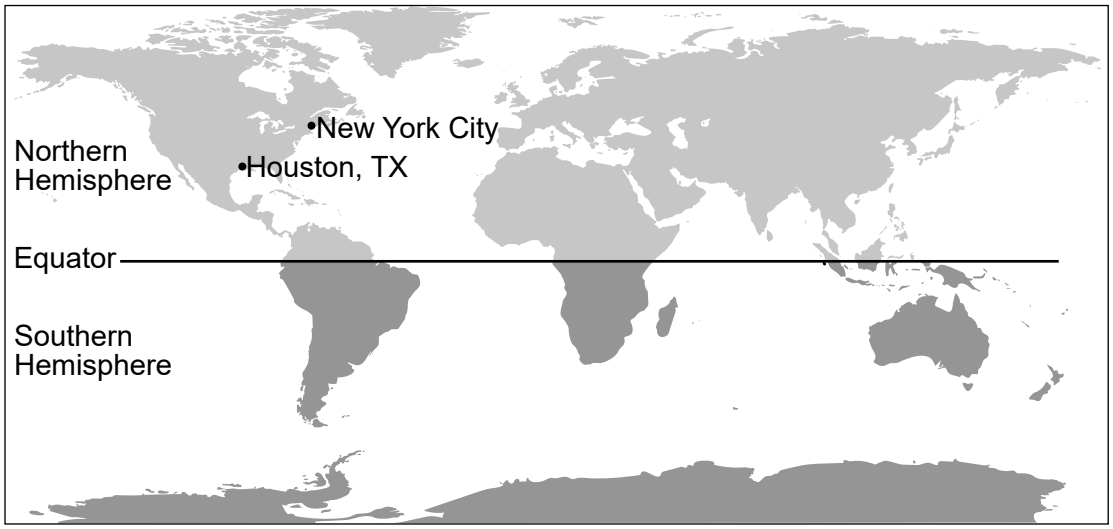
C In the spring, the monthly surface water temperature is at its highest, supplying energy to help develop storms into hurricanes.

D In the spring, the monthly precipitation near Houston is at its highest, supplying energy to help develop storms into hurricanes.

GO ON

The map below shows the location of Houston, Texas and New York City.

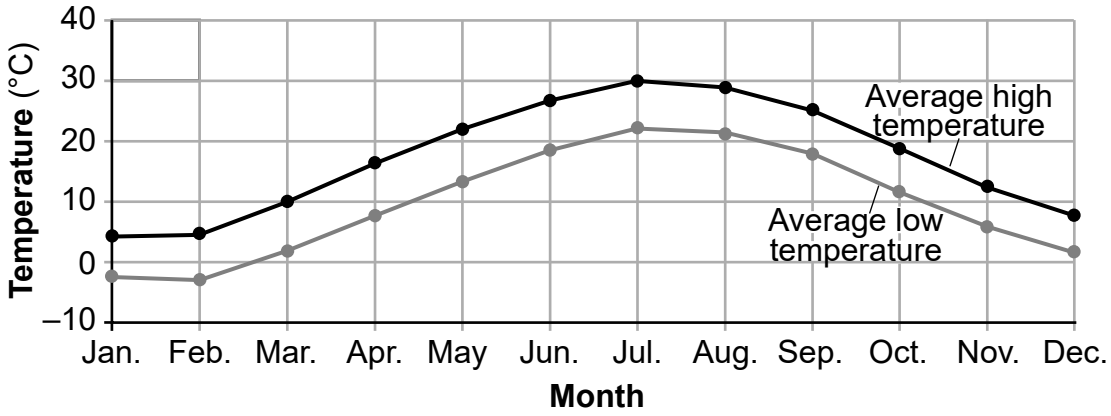
World Map



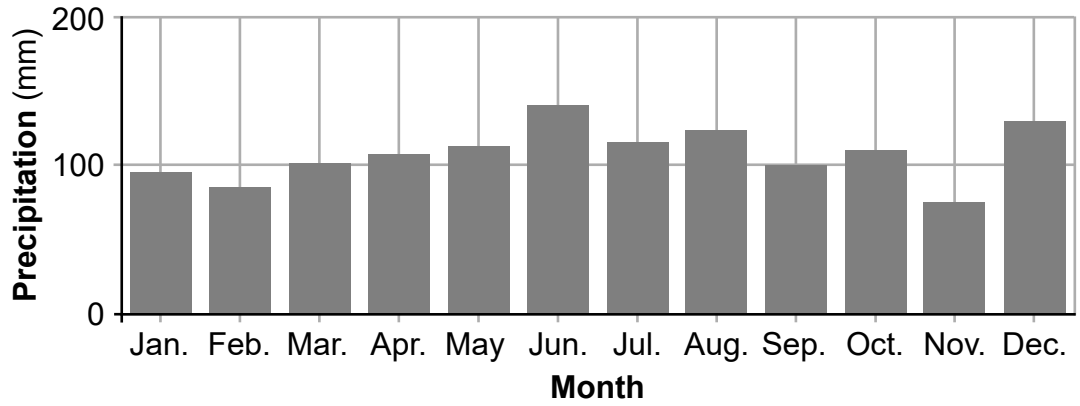
GO ON

The graphs and table below show some information about the climate of New York City.

Monthly Average High and Low Air Temperatures in New York City



Monthly Average Precipitation in New York City



Monthly Average Surface Water Temperature of Atlantic Ocean near New York City

Month	Jan.	Feb.	Mar.	Apr.	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.
Temperature (°F)	40.6	38.4	41.0	47.7	56.9	66.1	72.1	74.9	72.4	64.4	54.4	46.2

27 Based on the information in the map and graphs, which statement correctly compares the locations and climate conditions of Houston, Texas, to New York City?

A Houston is closer to the equator, has greater air temperatures in winter months, and more precipitation in December.

B Houston is closer to the equator, has greater air temperatures in summer months, and more precipitation in July.

C New York City is closer to the equator, has lower air temperatures in winter months, and less precipitation in July.

D New York City is closer to the equator, has lower air temperatures in summer months, and greater precipitation in December.

GO ON

Base your answers to questions 28 through 31 on the information below and on your knowledge of science.

Locust Trees

Locust trees are native to North America. They are found in various locations across the United States. There are many different types of locust trees, such as the honey locust. This tree gets its name from the honey-like substance found in its seed pods. The sweet seed pods of honey locusts are an important food source. White-tailed deer, rabbits, squirrels, and foxes will eat the honey locust seed pods. Deer will also eat the bark of the honey locust tree in the winter when other food sources are scarce.

Honey Locust Tree



Seed Pod of a Honey Locust Tree



28 Explain how the taste of the honey locust seed pods contributes to the reproduction of the honey locust trees. [1]

Many mature honey locust trees have long thorns on their trunks and smaller thorns on their branches. Not all honey locust trees have thorns. The thornless variety of this tree has similar seed pods, leaves, and flowers to the variety with thorns. Honey locust trees are often planted in cities due to their ability to survive in poor air quality and periods of dryness. Seeds from honey locust trees with thorns may grow into trees without thorns. Likewise, seeds from honey locust trees without thorns may grow into trees with thorns.

Trunk of Honey Locust Tree without Thorns



Trunk of Honey Locust Tree with Thorns



29 Which statement describes a variation in a trait that exists in honey locust trees?

A Honey locust trees are often planted in cities.

B All honey locust trees have seed pods.

C Some honey locust trees have thorns.

D Honey locust trees can live in areas with poor air quality.

30 Which argument supports the claim that honey locust trees with thorns are more likely to survive than honey locust trees without thorns?

A The thorns of these trees will produce more food for the tree.

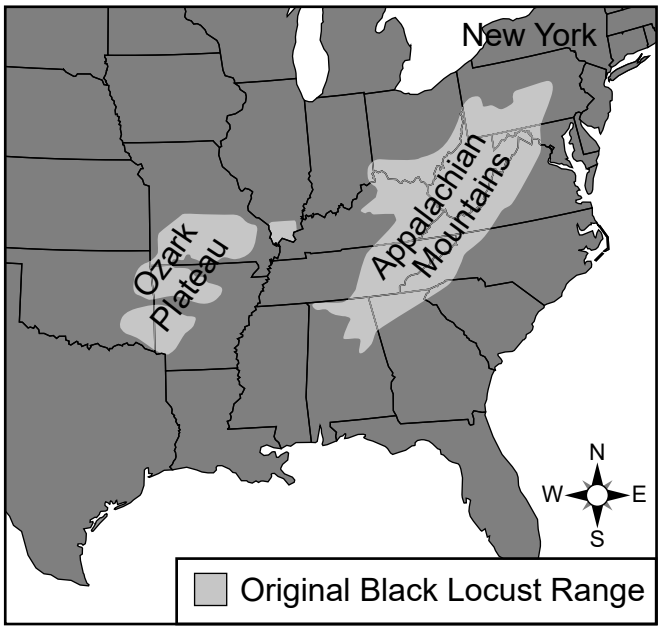
B The thorns of these trees help absorb more water for growth.

C The thorns of these trees provide habitats for moths and caterpillars.

D The thorns of these trees make it difficult to eat the tree's bark.

The black locust is a tree similar to the honey locust. The map below shows the original range for the black locust in the United States. This range was primarily in the Appalachian Mountains and the Ozark Plateau.

Original Black Locust Range



The distribution of the black locust has now spread in the United States to the Midwest and Northeast, including New York State. This tree has a fast growth rate and grows in groups. These groups decrease land surface temperatures. The groups also decrease the amount of light that can reach low-growing plants of grasslands and prairies.

31 Which set of terms correctly completes the passage below?

The introduction of the black locust to the Midwest prairie ecosystem could __1__ the amount of grass and clover, resulting in a(n) __2__ in the number of grasshoppers and owls because the black locust could __3__ the balance of the ecosystem.

A

1	increase
2	decrease
3	upset

C

1	decrease
2	increase
3	improve

B

1	decrease
2	decrease
3	upset

D

1	increase
2	increase
3	improve

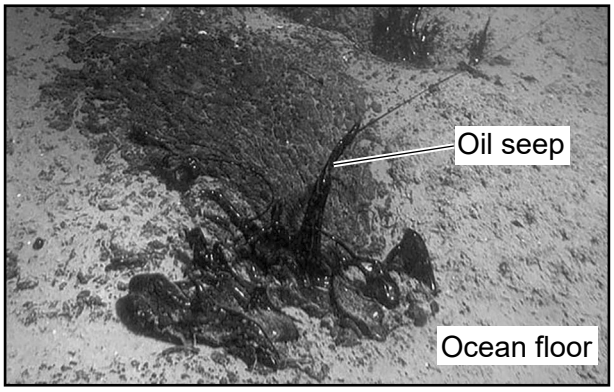
GO ON

Base your answers to questions 32 through 35 on the information below and on your knowledge of science.

Natural Energy Resources

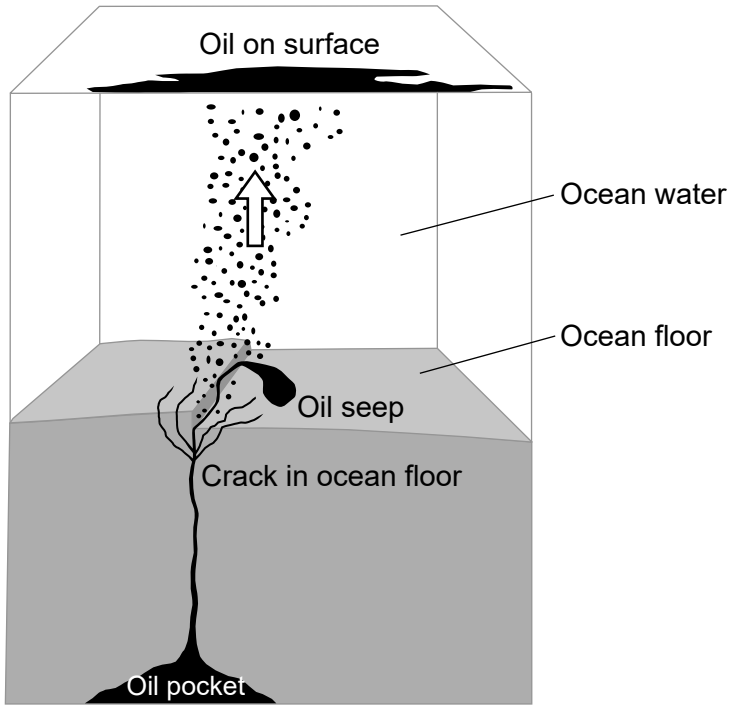
Humans use many types of natural resources to provide energy for their daily lives. Petroleum, also called crude oil or oil, is a fossil fuel formed millions of years ago. It is an important fuel source used around the world. Oil can be refined to produce heating fuel and gasoline. It can also be used to produce electrical energy.

Oil is found in pockets under Earth's surface. The photograph below shows an oil seep. Oil seeps occur when crude oil leaks through natural cracks in the ocean floor.



The model below represents the positions of both the oil pocket and oil seep in relation to the ocean floor. The direction of motion of oil droplets in the ocean is indicated.

Model of Movement of Oil in an Oil Seep



32 Which two Earth spheres are shown interacting in the model as a result of this oil seep?

A hydrosphere and atmosphere

B biosphere and geosphere

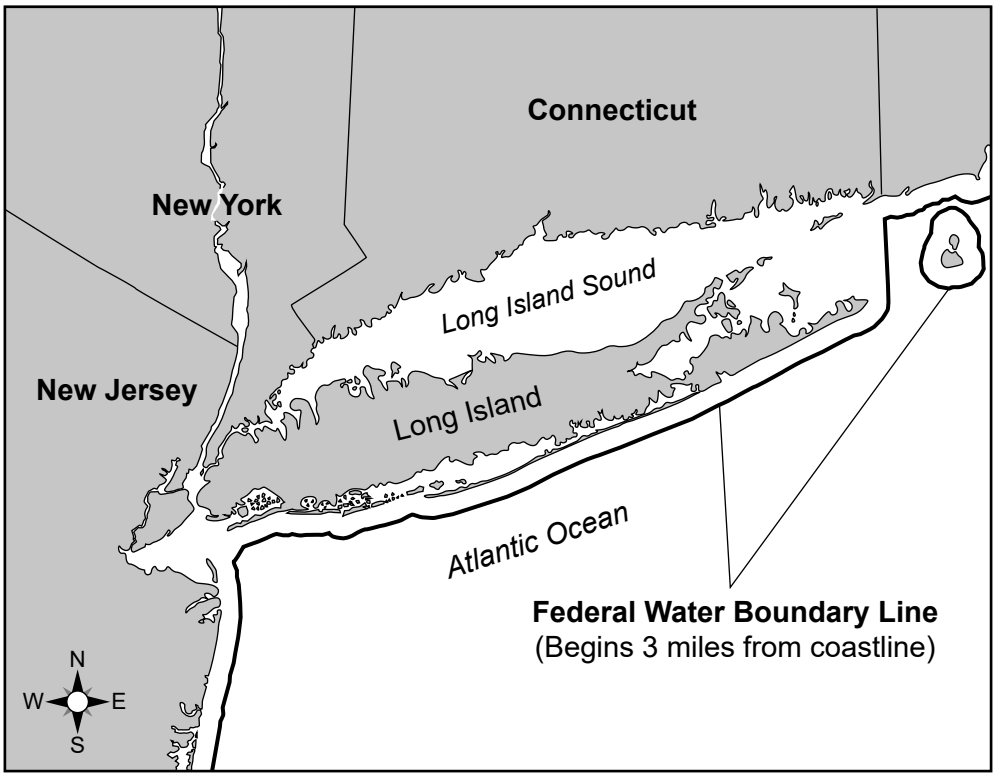
C geosphere and hydrosphere

D atmosphere and biosphere

GO ON

Offshore drilling is a way of obtaining underground oil from Earth by drilling into the ocean floor where oil pockets have been found. In 2019, New York State passed a law banning offshore drilling in the ocean floor within three miles of the state's shoreline.

New York's Marine and Coastal District Waters



33 Offshore drilling can have negative effects on the environment. Describe **one *negative*** effect that may have led New York State to ban offshore drilling. [1]

GO ON

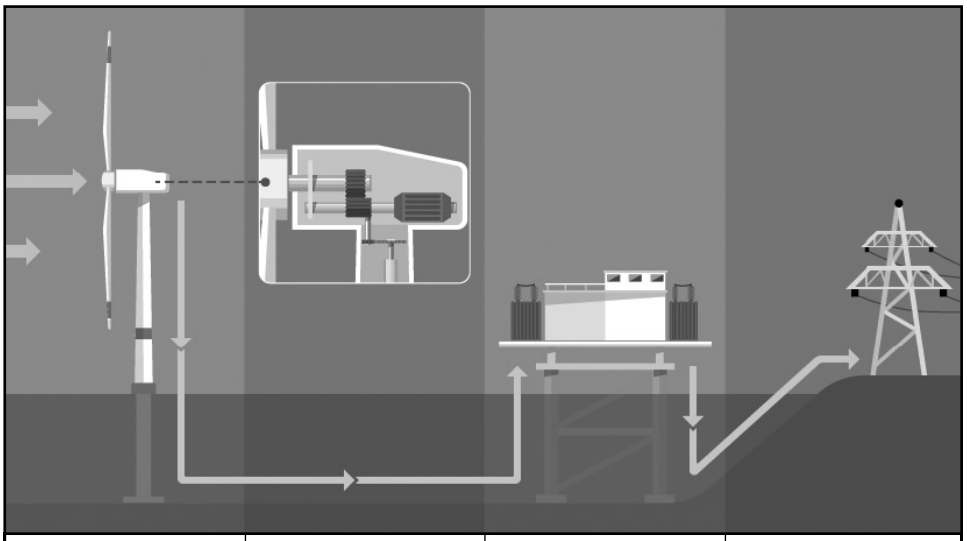
Wind is another natural resource that humans use to provide energy for their daily lives. Wind is used to produce electricity with large windmills called wind turbines. A wind farm is a group of wind turbines in the same location. The photograph below shows Maple Ridge Wind Farm, located in Lewis County, New York.



34 Describe how the Maple Ridge Wind Farm helps protect the environment. [1]

The model below shows how a wind turbine functions to produce electrical energy.

Model of How a Wind Turbine Works



1	2	3	4
Wind turns blades of the wind turbine.	Turbines turn gears in generators, making electrical energy.	Transformers at substations store electrical energy for use.	Electrical energy joins power grid and is distributed.

35 What type of energy is the wind turbine converting into electrical energy?

A light

B heat

C motion

D sound

STOP